

Test 21: ?* spinorial phase ? ?/2

Verdict: PASS | Generated: 2026-09-02 15:31:19 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

EXACT geometric phase check.

METHOD: EXACT geometric phase check.

Test 21 HARDCORE: 256-bit a_C err 1.23e-19, b_C err 6.39e-17, arg dev 0.1259°, ?±1e-6 worst dev 0.1259°.

Result Test 21b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

- `plots/plot\_01` (.png 600dpi / .svg / .pdf)

Plots

- `plots/animation.gif` ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v19.jl --test 21 --no-two-pass

Independent verification guide: [recompute logs](#) (recompute computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt).

(no log_comp entries)

FULL COMPUTATION LOG

??

CONSOLE CAPTURE

TEST 21 (pass 2, HARDCORE): ? spinorial phase ? 256-bit precision audit

??

?_C = ?/7, b?(K3) = 22 (256-bit recomputation)

a_C: Float64 8.276305e-04 vs 256-bit 8.276305e-04 ? |?| = 1.23e-19

b_C: Float64 0.376510 vs 256-bit 0.376510 ? |?| = 6.39e-17

arg(?*) at 256-bit: 89.874055° (deviation from 90°: 0.1259°)

? = ?/7-1·1e-6: arg(?*) deviation = 0.1259°

? = ?/7+1·1e-6: arg(?*) deviation = 0.1259°

??

Float64 round-off PASS ? |?a|=1.23e-19, |?b|=6.39e-17 (machine-? scale)

arg(?*) ? ?/2 (256-bit) PASS ? deviation 0.1259° < 1°

?_C ± 1e-6 stability PASS ? worst deviation 0.1259° across the ±1e-6 scan

Test 21b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS